Abdelrahman Abouroumia

AI & Backend Engineer

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Software engineer building backend services and AI systems. Builds Python microservices at Tutanic. Co-founded Zaylon AI and built its AI sales agent and the backend behind it.

Experience

Software Engineer, AI Systems Sep 2026 to Present
Tutanic | *Backend services and AI systems.*
- Build backend microservices in Python (FastAPI, Django) that talk to each other through NATS JetStream.
- Build AI features with LangGraph on AWS Bedrock, and deploy services to Kubernetes using Helm.

Software Engineer & Co-Founder May 2025 to Present
Zaylon AI | *AI sales assistant for online stores on WhatsApp, Instagram and web chat*
- Led engineering and designed the backend: Python microservices that communicate through Kafka.
- Built the AI agent (LangGraph) that answers shoppers, finds products and completes orders inside the chat.
- Made the agent understand and reply in Egyptian Arabic and Franco-Arabic (Arabic typed in Latin letters).
- Connected 6 e-commerce platforms, including Shopify and WooCommerce, and 3 payment providers, including Stripe.
- Kept every store's data fully separate from other stores, with automated checks that block unsafe code.

AI Coding Expert Sep 2024 to Feb 2026
Scale AI | *Contract*
- Reviewed and ranked AI-generated code for 18 months, used to train leading AI models.
- Wrote tests to decide whether each code answer was correct, efficient and handled edge cases.

Software Integration Engineering Intern Jul 2025
Ejada
- Built banking REST APIs in Spring Boot; sped up slow database queries and added Redis caching.
- Used Kafka to pass messages between services, and secured the API gateway with OAuth2 and rate limits.

Automotive Cybersecurity Intern Aug 2024 to Oct 2024
Swift-ACT
- Mapped security threats to in-car networks (CAN bus) and simulated attacks on them in CARLA and CANoe.

Projects

AdvisoryDrift: AI Agent That Corrects Security Advisories *Award winner* | GitHub | Case Study
- Built an AI agent that finds the right affected versions for published security alerts by reading git history.
- Tested on real past corrections, it got 15 of 38 fully right, three times as many as a scripted approach.

Talos: Team Chat App with an AI Assistant *Graduation project* | GitHub | Case Study
- Built the AI assistant that answers questions from the team's own documents and shows where each answer came from.
- Built tests that score answer quality, and used them to find and fix why early answers were weak.

Virtual Banking Microservices GitHub | Case Study
- Built a banking system from Spring Boot microservices, where money transfers are processed through Kafka.
- Made sure a transfer either completes fully or rolls back fully, and can be traced end to end.

commentdraft *Published package* | PyPI | GitHub | Case Study
- Python tool that drafts replies to social media comments from your own notes; nothing is sent without approval.

Education

Alexandria University, Faculty of Engineering Graduated Jul 2026
B.Sc. in Computer & Communications Engineering, AI Concentration | GPA **3.75 / 4.0**

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C/C++
Backend: FastAPI, Django, Spring Boot, Microservices, Event-Driven Architecture, Kafka, NATS JetStream, REST APIs
Data: PostgreSQL, PgBouncer, pgvector, Redis, Dragonfly, Milvus, Amazon S3
AI: LangGraph, LangChain, MCP, CopilotKit, AWS Bedrock, RAG, LLM Agents, Agent Evaluation, PyTorch, NLP
Infrastructure: Kubernetes, Helm, Docker, AWS, Railway, GitHub Actions, CI/CD, Linux, pytest, Testcontainers

Additional

Award: Best Demonstrated Improvement, micro1 Frontier Engineering Challenge 2026
Certifications: micro1 Certified Software Engineer | Google AI Agents | Meta Back-end | AWS Cloud & ML Foundations
Competitions: MTC-AIC3 BCI (Egypt National AI) | ECPC (3x) | Hepsiburada NLP Hackathon (TEKNOFEST)
Spoken languages: English (Fluent) | Arabic (Native) | German (A1) | Spanish (Conversational)